$$\beta(s) = \sum_{k \geq 0} (-1)^k (2k+1)^{-s}$$

$\beta(1) = \pi/4$
Known evaluation

$\beta(3) = \pi^3/32$
Known evaluation

$\beta(2) = G$
No proof
established here

Exact identities at neighbouring arguments do not decide β(2).